

HydroFlow RP9

Multi-Frequency Acoustic Doppler Current Profiler



HydroFlow RP9

The HydroFlow RP9 is an advanced, multi-frequency Acoustic Doppler Current Profiler (ADCP) developed by Satlab. Designed for versatility and performance, The HydroFlow RP9 delivers high-quality river discharge and velocity profiling data across a wide range of environments. By integrating multiple acoustic frequencies, the RP9 provides robust and continuous measurements from shallow streams to deep channels, making it a comprehensive solution for hydrological and hydrographic applications.



Key Features

1 Intelligent Adaptive Operation

Automatically switches between NB, BB, and PC modes for optimal performance under varying flow and depth conditions.

2 Simultaneous Dual-Frequency Profiling

Simultaneous dual-frequency operation for detailed surface and extended bottom measurements.

3 Ultra-Shallow Water Measurement

Minimum cell size of 0.02 m and operation in water depths as shallow as 0.05 m for exceptional low-water performance.

4 Wide Environmental Adaptability

Reliable measurements from shallow streams to deep and fast-flowing rivers.

Installation Method

① Manned Vessel

- Large river discharge measurement
- Long-duration monitoring operations
- Offshore or wide-area cross-section surveys



② Unpowered Float

- Low-velocity river flow measurement
- Bridge-based ADCP surveys
- Cableway-supported hydrological measurements



③ Unmanned Surface Vessel

- Flood discharge measurement in hazardous areas
- Shallow water and inaccessible river sections
- Rapid deployment emergency surveys



Discharge Measurement Solution



HydroBoat 1200

- Integrated position and heading
- Autonomous navigation (hovering, obstacle avoidance, return)
- Automatic two-point transect survey
- Expandable sensors for water quality & sampling
- Quick deployment, instant measurement readiness



Trimaran & Njord Plus

- Integrated position, heading and data transmission module
- Same Android-based data acquisition software as USV
- Cost-effective survey solution
- Simple setup and easy field operation

Specification

Model		HydroFlow RP9		
Frequency		4*3.0MHz, 4*1.2MHz, 1*0.6MHz		
Working mode		Broadband, Narrowband Pulse Coherent		
Cell Size		0.02-4m		
Cell Number		1-260		
Ping Rate		20Hz		
Velocity Profiling	Range	±5m/s (Typical), ±20m/s (Max)		
	Accuracy	±0.25% ±0.2cm/s		
	Resolution	1mm/s		
	Profile range	0.05-40m		
Bottom Track Range		0.05m~55m		
Depth Range/Accuracy/Resolution		0.1m~120m	1%	1%
Temperature Range/Accuracy/Resolution		30°C~95°C	±0.1°C	0.01°C
Compass Range/Accuracy/Resolution		0°~360°	±0.1°	0.01°
Motion Range/Accuracy/Resolution		±90°	±0.15°	0.01°
Data communication		RS-232, Bluetooth 200m, RS-422 (optional), RS-485 (optional), 100M Ethernet (optional)		
Rating		50m		
Memory		32GB		
Power		10-36VDC		
Material		PP		
Weight		2.46kg (air)		
Dimension		Φ129mm * 195mm		
Working/Storage temperature		-5°C~+50°C		-30°C~70°C



*Specifications are subject to change without notice. We reserve the right of final interpretation.
*All listed parameters are either theoretical values or measured under controlled conditions.



Headquarters:

Geosolution i Göteborg AB
Stora Ävägen 21, 436 34 ASKIM,
Sweden

Regional Offices:

Budapest, Hungary
Ankara, Turkey
Dubai, UAE
New Delhi, India
Scottsdale, USA
Tokyo, Japan
Hong Kong, China

www.satlabgeo.com